

2- Bayesian Inference (i)

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Overview

- 1- Poisson distribution
- 2- Conjugate distributions
- 3- Prior and Posterior predictive distributions.
- 4- Inference for Normal distribution, σ^2 known and μ , σ^2 unknown.

1-Poisson distribution

The probability function for a single observation coming from Poisson distribution

is: $P(x|\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$ for $x=0,1,2,\dots$ and $\lambda>0$. Moreover, $E(X)=\text{Var}(X)=\lambda$.

Assuming that we have a series of observations $x_1, \dots, x_n$ independent and identically distributed (iid), the joint probability function is the product of the individual probabilities, and the likelihood function is,

$$L(\lambda|x_1, \dots, x_n) \propto P(x_1, \dots, x_n|\lambda) \\ \propto \lambda^{\sum_{i=1}^n x_i} e^{-n\lambda}$$

Example: The following data (Sokal and Rohlf, 1981, p. 84) refer to the distribution of yeast cells in 400 squares of a hemacytometer, a counting chamber used in making counts of blood cells and other microscopic structures suspended in liquid.

x	0	1	2	3	4	5	6	7	8	9	Total
n_x	75	103	121	54	30	13	2	1	0	1	400

Here the likelihood function is, $L(X|\lambda) \propto \lambda^{720} e^{-400\lambda}$ and the MLE is $\hat{\lambda} = 1.8$ (just the mean)

Prior densities

-Noninformative (or *objective*): A simple way is to consider as a prior, an uniform distribution. However the parameter has an infinite range and this might be a problem because the area under the curve must be 1. Then, this kind of prior is an *improper* distribution. For many statisticians this is not a problem if the posterior distribution is *proper*. This improper density will be denoted as $\pi_1(\lambda) \sim 1$.

Another option is to choose an uniform distribution over a finite range, for instance, $[\min(x_1 \dots x_n), \max(x_1 \dots x_n)]$. This density will be denoted as $\pi_2(\lambda)$.

-Informative (or *subjective*): A flexible continuous distribution over the positive reals is the Gamma(α , β) distribution. This density, $\pi_3(\lambda)$, depends of 2 parameters:

$$\pi_3(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}, \quad \lambda > 0$$

$$E(X) = \frac{\alpha}{\beta}, \quad Var(X) = \frac{\alpha}{\beta^2}$$

The parameters (*hyper-parameters*) of the prior Gamma distribution can be estimated from the original data. The posterior density has the form,

$$\begin{aligned} f(\lambda|X) &\propto \pi_3(\lambda)L(X|\lambda) \propto \lambda^{\alpha-1}e^{-\beta\lambda}\lambda^{\sum_{i=1}^n x_i}e^{-n\lambda} \\ &\propto \lambda^{\alpha+\sum_{i=1}^n x_i-1}e^{-(\beta+n)\lambda} \end{aligned}$$

and consequently, the posterior distribution is also Gamma ($\alpha + \sum_{i=1}^n x_i$, $\beta + n$).

Taking the uniform prior density $\pi_1(\lambda)$, that is, an improper uniform over $[0, \infty]$, the following posterior density is obtained: $f_1(\lambda | X) = c \lambda^{720} e^{-400\lambda}$, for $0 < \lambda$.

The posterior distribution is just a Gamma(721,400).

However, using the proper uniform prior density $\pi_2(\lambda)$, the posterior is not Gamma distributed because this is defined over $0 \leq \lambda \leq 9$. It can be shown that $f_2(\lambda | X)$ is just the density of a Gamma(721,400) divided by its area over $[0,9]$. However this area is almost 1 and consequently $f_1(\lambda | X) \approx f_2(\lambda | X)$.

In order to use the informative prior $\pi_3(\lambda)$ we need to estimate the hyper-parameters. It can be done applying the method of the moments to the original data:

$$E(X) = \frac{\alpha}{\beta} = \bar{x} = 1.8 \quad , \quad Var(X) = \frac{\alpha}{\beta^2} = s^2 = 1.9649$$

$$\hat{\alpha} = \frac{\bar{x}^2}{s^2} = 1.649 \quad , \quad \hat{\beta} = \frac{\bar{x}}{s^2} = 0.916$$

Then the posterior density $f_3(\lambda | X) = c \lambda^{1.649+720-1} e^{-(0.916+400)\lambda}$ is a Gamma(721.649, 400.916).

Note that, because we are assuming that data comes from a Poisson distribution where $E(X) = Var(X)$, it is reasonable to use as a prior a Gamma($\alpha, 1$). Then, the hyper-parameter can be estimated by the sample mean, obtaining a posterior density $f_4(\lambda | X)$, that is, a Gamma(721.8, 401).

Using each posterior, we obtain the following point estimates of the parameter:

	$f_1 \approx f_2$	f_3	f_4	
1- Expectation	1.8025	1.8000	1.8000	α/β
2- Mode	1.8	1.7975	1.7975	$(\alpha-1)/\beta$
3- Median	1.8017*	1.7992	1.7992	
4- Classical (non-Bayesian) :	1.8			

(*): `qgamma(0.5,721,400)`

95% credibility intervals are calculated using the corresponding percentiles:

$f_1 \approx f_2$	f_3	f_4
[1.673 , 1.936]*	[1.671 , 1.934]	[1.671 , 1.934]

(*): the extremes are `qgamma(0.025,721,400)` and `qgamma(0.975,721,400)`

2-Conjugate distributions

In Bayesian language, if the posterior distributions are in the same family as the prior probability distribution, the prior and posterior are then called *conjugate distributions*, and the prior is called a conjugate prior for the likelihood.

Table 2.5. Common conjugate priors

Sampling Distribution (Parameter)	Conjugate Prior
Binomial (π)	Beta
Exponential (λ)	Gamma
Gamma (λ)	Gamma
Multinomial (π)	Dirichlet
Multivariate Normal (μ, Σ)	Normal Inverse Wishart
Negative Binomial (π)	Beta
Normal (μ, σ^2 known)	Normal
Normal (σ^2, μ known)	Inverse Gamma
Normal (μ, σ^2)	Normal Inverse Gamma
Pareto (β)	Gamma
Poisson (λ)	Gamma
Uniform($0, \beta$)	Pareto

From the book “Bayesian Reliability” by Michael Hamada

3a-Predictive prior distribution

We assume that our observations follow a certain distribution (model) whose density (or probability function) is $p(x|\theta)$. On the other hand, we also have the prior density $\pi(\theta)$ which represents the uncertainty in θ before taking the sample.

So for some new data value x_{new} , averaging over $\pi(\theta)$ gives the so called *prior predictive distribution*. Therefore, the prior predictive density (or probability function) is computed as follows:

$$p(x_{\text{new}}) = \int_{\Theta} p(x_{\text{new}}|\theta)\pi(\theta)d\theta$$

Here Θ indicates the domain of the parameter.

When parameter θ is a vector, the integral is multiple.

The prior predictive distribution is a kind of *continuous mixture* of the distribution of the data (the observations).

Example 1: In the example of the distribution of the yeast cells,

$$p(x_w|\lambda) = \frac{e^{-\lambda} \lambda^{x_w}}{x_w!}$$

(Here we write x_w instead of x_{new} to make easy the notation!)

Therefore, considering the prior $\pi(\lambda) \approx \text{Gamma}(1.65, 0.92)$, the prior predictive distribution is,

$$\begin{aligned} p(x_w) &= \int_0^\infty \frac{e^{-\lambda} \lambda^{x_w}}{x_w!} \frac{0.92^{1.65}}{\Gamma(1.65)} \lambda^{1.65-1} e^{-0.92\lambda} d\lambda = \frac{0.92^{1.65}}{x_w! \Gamma(1.65)} \int_0^\infty \lambda^{x_w+1.65-1} e^{-1.92\lambda} d\lambda \\ &= \frac{0.92^{1.65}}{x_w! \Gamma(1.65)} \frac{\Gamma(x_w+1.65)}{1.92^{x_w+1.65}} = \frac{0.3299925 \Gamma(x_w+1.65)}{1.92^{x_w} x_w!} \end{aligned}$$

Then, the probabilities of $x_w = 0, 1, 2, 3$ are 0.29703, 0.25526, 0.17616, 0.11163, respectively. This count distribution is a special case of the so called *negative binomial* distribution.

Example 2: Let x be the number of heads in 20 independent throws of a coin. The probability of “head” is θ . We assume that the coin is “almost” fair, that is $E(\theta)=1/2$ and $P(0.4<\theta<0.6) = 0.95$, considering a suitable beta prior distribution.

- What is the predictive prior distribution?
- What is the probability to get 10 heads in 20 throws?
- What is the probability to get 1 head in 1 throw?

First of all, the suitable beta distribution is such that, $\frac{\alpha}{\alpha+\beta} = 0.5$

It implies that $\alpha = \beta$

Then, we have to solve the equation (R notation) $\text{pbeta}(0.6,a,a)-\text{pbeta}(0.4,a,a)=0.95$

This is the script for doing this:

```
uniroot(function(a){pbeta(.6,a,a)-pbeta(.4,a,a)-0.95},c(1,100))
```

```
$root
```

```
[1] 47.29982
```

```
$f.root
```

```
[1] -1.824912e-09
```

Then, here $p(x_w|\theta) = \frac{20!}{(20-x_w)!x_w!} \theta^{x_w} (1-\theta)^{20-x_w}$

and the prior density is $\pi(\theta) = \frac{\Gamma(94.6)}{\Gamma(47.3)\Gamma(47.3)} \theta^{47.3-1} (1-\theta)^{47.3-1}$

a) The predictive prior density is

$$\begin{aligned} p(x_w) &= \frac{\Gamma(94.6)20!}{\Gamma(47.3)\Gamma(47.3)(20-x_w)!x_w!} \int_0^1 \theta^{47.3+x_w-1} (1-\theta)^{67.3-x_w-1} d\theta \\ &= \frac{\Gamma(94.6)20!\Gamma(47.3+x_w)\Gamma(67.3-x_w)}{\Gamma(47.3)^2(20-x_w)!x_w!\Gamma(114.6)} \end{aligned}$$

This is a particular case of the so called *beta-binomial* distribution.

b) The probability to get 10 heads in 20 throws is,

$$p(10) = \frac{\Gamma(94.6)20!\Gamma(57.3)^2}{\Gamma(47.3)^2(10!)^2\Gamma(114.6)} = 0.1600$$

This is the script in R to avoid overflow:

```
exp(lgamma(94.6)+lgamma(21)+2*lgamma(57.3)-2*lgamma(11)-lgamma(114.6)-
2*lgamma(47.3))
```

The probability to get 10 heads in 20 throws without uncertainty, that is, with a degenerate prior concentrated in 0.5, is $\text{dbinom}(10,20,.5)= 0.1762$

c) The probability to get 1 head in 1 throw is

$$p(1) = \frac{\Gamma(94.6)}{\Gamma(47.3)\Gamma(47.3)} \int_0^1 p(1|\theta) \theta^{47.3-1} (1-\theta)^{47.3-1} d\theta$$

But now, $p(x_w|\theta) = \theta^{x_w} (1-\theta)^{1-x_w}$ and $p(1|\theta) = \theta$

Therefore, $p(1)$ is the expectation of a $\text{Beta}(47.3,47.3)$, that is $p(1)=0.5$

3b-Predictive posterior distribution

After taking the observations, we have a better representation of the uncertainty in θ by means of our posterior $f(\theta|X)$. So the posterior predictive distribution for a new data

point or vector X_{new} is,

$$p(X_{\text{new}}|X) = \int_{\Theta} p(X_{\text{new}}|\theta) f(\theta|X) d\theta$$

Again Θ indicates the domain of the parameter.

Again, when parameter θ is a vector, the integral is multiple.

Similarly, the posterior predictive distribution is a kind of *continuous mixture* of the distribution of the data (the observations).

Example 1 (continuation): In the example of the yeast cells,

$$p(x_w|\lambda) = \frac{e^{-\lambda} \lambda^{x_w}}{x_w!}$$

(Here we write x_w instead of x_{new} to make easy the notation!)

Considering the prior $\pi(\lambda) \approx \text{Gamma}(1.65, 0.92)$, and the 400 observations, we got a posterior $f(\lambda|X) \approx \text{Gamma}(721.65, 400.92)$. Then, the posterior predictive distribution for one future observation is,

$$\begin{aligned} p(x_w|X) &= \int_0^\infty \frac{e^{-\lambda} \lambda^{x_w}}{x_w!} \frac{400.92^{721.65}}{\Gamma(721.65)} \lambda^{721.65-1} e^{-400.92\lambda} d\lambda \\ &= \frac{400.92^{721.65}}{x_w! \Gamma(721.65)} \frac{\Gamma(x_w + 721.65)}{401.92^{x_w + 721.65}} = \frac{0.1656722 \Gamma(x_w + 721.65)}{401.92^{x_w} x_w! \Gamma(721.65)} \end{aligned}$$

Then, the probabilities of $x_w = 0, 1, 2, 3$ are 0.16567, 0.29747, 0.26742, 0.16050, respectively. This is again a special case of the so called *negative binomial* distribution. To avoid overflows, this is the script for computing the probabilities:

`exp(log(.1656722)+lgamma(x+721.65)-x*log(401.92)-lgamma(x+1)-lgamma(721.65))`

Example 2 (continuation): We have thrown the coin 100 times obtaining 43 heads.

- What is the predictive posterior distribution for the number of heads in a future 20 independent throws of a coin?
- What is now the probability to get 10 heads in 20 throws?
- What is the probability to get 1 head in a future throw?

The prior distribution is $\pi(\theta) \approx \text{Beta}(47.3, 47.3)$, the likelihood is $L(X|\theta) = \theta^{43}(1-\theta)^{57}$ so that the posterior density is, $f(\theta | X) = c \theta^{90.3-1}(1-\theta)^{104.3-1}$, thus the posterior distribution is a $\text{Beta}(90.3, 104.3)$.

a) The predictive posterior distribution for the number of heads in 20 independent throws of a coin is:

Again, $p(x_w | \theta) = \frac{20!}{(20-x_w)!x_w!} \theta^{x_w} (1-\theta)^{20-x_w}$, therefore

$$\begin{aligned} p(x_w | X) &= \frac{\Gamma(194.6)20!}{\Gamma(90.3)\Gamma(104.3)(20-x_w)!x_w!} \int_0^1 \theta^{90.3+x_w-1} (1-\theta)^{124.3-x_w-1} d\theta \\ &= \frac{\Gamma(194.6)20!\Gamma(90.3+x_w)\Gamma(124.3-x_w)}{\Gamma(90.3)\Gamma(104.3)(20-x_w)!x_w!\Gamma(214.6)} \end{aligned}$$

The probability function is plotted using the following script:

```
x=seq(0,20,1)
y=exp(lgamma(194.6)+lgamma(21)+lgamma(90.3+x)+lgamma(124.3-x)-lgamma(x+1)-
lgamma(21-x)-lgamma(90.3)-lgamma(104.3)-lgamma(214.6))
plot(x,y)
```

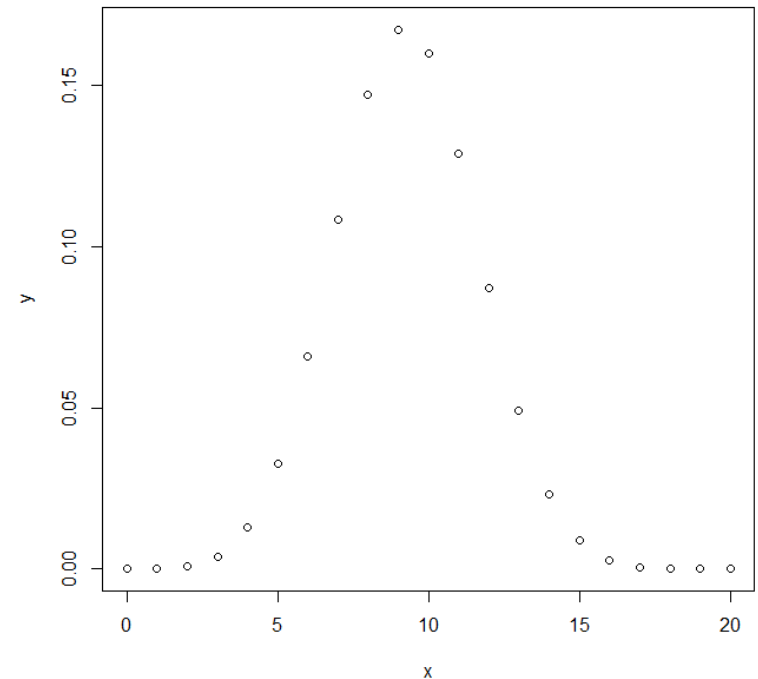
In particular,

b) the probability to get 10 heads in 20 throws is

$$p(10|X) = 0.15998$$

c) The probability to get 1 head in 1 throw is

similarly, the expectation of a Beta(90.3,104.3),
that is, $p(1|X)=0.464$



Exercises:

1- Let y be the number of heads in n spins of a coin, whose probability of heads is θ .

a) If your prior distribution for θ is uniform on $[0, 1]$, derive your prior predictive distribution for y_{new} , $P(y_{\text{new}}=k)$, $k=0,1,\dots,n$.

b) Suppose you assign a $\text{Beta}(\alpha, \beta)$ prior distribution for θ , and then you observe y heads out of n spins. Show algebraically that your posterior mean of θ always lies between your prior mean, $\alpha/(\alpha+\beta)$, and the observed relative frequency of heads, y/n .

c) Show that, if the prior distribution on θ is uniform, the posterior variance of θ is always less than the prior variance. Give an example of a $\text{Beta}(\alpha, \beta)$ prior distribution and data y, n , in which the posterior variance of θ is higher than the prior variance.

(Bayesian Data Analysis, Gelman et al. 2014)

2-We are analysing the lifetime of a bacterium at a certain temperature, which follows an Exponential distribution with density $f(x|\mu) = \exp(-x/\mu)/\mu$, where parameter μ is the mean life expectancy.

a) Compute the prior predictive distribution of an observation using an Inverse Gamma density as a prior for the parameter μ with population mean and standard deviation of 4.5 and 0.5 respectively. Plot the predictive density and compute its expectation and variance.

Exercises:

- b) Using a Uniform[3.63,5.37] as a prior for the parameter μ , simulate values of the predictive prior distribution of a new observation. Plot the approximate predictive density and compute its approximate expectation and variance. Compare with the results of section a).
- c) We observe the lifetime of $n=8$ bacteria (expressed in hours): 2.56, 0.42, 3.72, 0.64, 10.7, 1.57, 9.62, 2.54. Using the prior considered in section a), compute the predictive posterior distribution of a future observation. Plot the predictive density and compute its expectation and variance.
- d) Simulate values of the predictive posterior distribution of a new observation when the prior is a Uniform[3.63,5.37]. Plot the approximate posterior predictive density and compute its approximate expectation and variance. Compare with the results of c).

4a- Normal distribution (with σ^2 known)

Suppose that we have n observations $x_1, \dots, x_n$ independent and identically distributed (iid), coming from a Normal distribution. The likelihood function is,

$$\begin{aligned} L(X|\mu, \sigma^2) &= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right) \\ L(X|\mu, \sigma^2) &= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{n}{2\sigma^2} (s^2 + (\bar{x} - \mu)^2)\right) \\ &\propto \frac{1}{\sigma^2}^{n/2} \exp\left(-\frac{n}{2\sigma^2} ((\bar{x} - \mu)^2)\right) \exp\left(-\frac{ns^2}{2\sigma^2}\right) \end{aligned}$$

(check this!)



Note how all the information of the sample is summarized in the sample mean and variance (they are sufficient statistics).

Suppose that σ^2 is known

If σ^2 is constant, the likelihood function can be written as, $L(X|\mu) \propto \exp\left(-\frac{n}{2\sigma^2} (\bar{x} - \mu)^2\right)$

-Noninformative prior: Improper density $\pi_1(\mu) \sim 1$.

-Informative prior: Conjugated density $\pi_2(\mu)$, $N(\mu_0, \sigma_0^2)$, (here μ_0, σ_0^2 are the hyper-parameters).

Posterior densities

- Taking the uniform prior density $\pi_1(\mu)$, an improper uniform over $[-\infty, \infty]$, the posterior density is $f_1(\mu|X) \sim N(\bar{x}, \sigma^2/n)$

Consequently, classical and Bayesian inference are “the same”.



- Taking the conjugated prior density $\pi_2(\mu)$, since the product of two Gaussian densities is a Gaussian density, the posterior is,

(check this!) $f_2(\mu|X) \sim N\left(\frac{\frac{n\bar{x}}{\sigma^2} + \frac{\mu_0}{\sigma_0^2}}{\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}}, \frac{1}{\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}}\right)$



Note that the expectation is just a weighted mean. If the sample mean and variance are used in order to estimate μ_0 and σ_0^2 , the posterior density is,

$f_2(\mu|X) \sim N\left(\bar{x}, \frac{1}{\frac{n}{\sigma^2} + \frac{1}{s^2}}\right)$ For large samples f_1 and f_2 are very similar.

Example: The following data refer to the measures of weights (in g.) of the same object, performed with a precision balance: 2.64, 2.67, 2.65, 2.67, 2.65
The balance has been calibrated and its precision (sd) is $\sigma=0.01$. Estimate the true weigh.

- Using the noninformative prior (or the classical method), the results can be summarized as, $\mu = 2.656 \pm 0.00877$
The interpretation of this 95% confidence (credibility) interval is different for Bayesian or classical inference.

- With the conjugated prior, because $s = 0.0134$, the interval is $\mu = 2.656 \pm 0.00831$

Note that, for this prior, the credibility interval is always smaller than the classical.



4b- Normal distribution (μ and σ^2 unknown)

Suppose that μ and σ^2 are unknown.

Remember the likelihood function,

$$L(X|\mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{n}{2\sigma^2}(s^2 + (\bar{x} - \mu)^2)\right) \\ \propto \left(\frac{1}{\sigma^2}\right)^{-n/2} \exp\left(-\frac{n}{2\sigma^2}(\bar{x} - \mu)^2\right) \exp\left(-\frac{n s^2}{2\sigma^2}\right)$$

This is an example of a multi-parameter model. The Bayesian procedure is just the same, but involving bivariate densities. The posterior density will be constructed as,

$$f(\mu, \sigma^2|X) \sim \pi(\mu, \sigma^2)L(X|\mu, \sigma^2)$$

However, we would like to make inferences about the marginal distributions given by the densities $g(\mu|X)$ and $g(\sigma^2|X)$, rather than the bivariate distribution $f(\mu, \sigma^2|X)$. In order to construct the prior, we assume that both parameters are independent, that is, $\pi(\mu, \sigma^2) = h(\mu) w(\sigma^2)$, the product of two densities.

-Noninformative prior: A kind of noninformative prior can be constructed using the improper density $h(\mu) \sim 1$. But an uniform prior for σ^2 gives an improper posterior. However, an uniform prior on $\log(\sigma^2)$ is fine and consequently, $w(\sigma^2) \sim 1/\sigma^2$. Therefore, the prior joint density is $\pi(\mu, \sigma^2) \sim 1/\sigma^2$.

Posterior density

The posterior is
$$f(\mu, \sigma^2 | X) \propto \frac{1}{\sigma^{n+2}} \exp\left(-\frac{n}{2\sigma^2} (\bar{x} - \mu)^2\right) \exp\left(-\frac{n s^2}{2\sigma^2}\right)$$

We can write the posterior in terms of the conditional distribution of μ and the marginal distribution of σ^2 : $f(\mu, \sigma^2 | X) = f(\mu | \sigma^2, X) g(\sigma^2 | X)$

Note that $f(\mu | \sigma^2, X)$ is the same than the posterior density of μ when σ^2 is known and the uniform prior is used. Then, $f(\mu | \sigma^2, X) \sim N(\bar{x}, \sigma^2/n)$

Consequently, the marginal density of σ^2 is,
$$g(\sigma^2 | X) \sim (\sigma^2)^{-\left(\frac{n}{2} + \frac{1}{2}\right)} \exp\left(-\frac{n s^2}{2\sigma^2}\right)$$

This is a *scaled inverse- χ^2 density* with $n-1$ degrees of freedom and scale parameter s^2 . This is also an *inverse-gamma* with parameters $(n-1)/2$ and $(n-1) s^2/2$.

To obtain the marginal density of μ , we have to calculate, $g(\mu|X) \sim \int_0^\infty f(\mu, \sigma^2|X) d\sigma^2$

From here, it can be shown (difficult!) that $\frac{\mu - \bar{x}}{s/\sqrt{n}}$ follows a Student-T distribution with $n-1$ degrees of freedom, like in the classical approach.

Firefox

W Scaled-inverse-chi-squared distribution ...

Modelització Avançada

W en.wikipedia.org/wiki/Scaled-inverse-chi-squared_distribution

scaled inverse chi square

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Scaled-inverse-chi-squared distribution

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Unsourced material may be [challenged](#) and [removed](#). (December 2009)

The **scaled inverse chi-squared distribution** arises in [Bayesian statistics](#). The family of scaled inverse chi-squared distributions contains an extra scaling parameter compared to the [inverse-chi-squared distribution](#), and it is essentially the same family of distributions as the [inverse gamma distribution](#), but using a [parametrization](#) that may be more convenient for Bayesian statistics. Specifically, the scaled inverse chi-squared distribution can be used as a [conjugate prior](#) for the [variance](#) parameter of a [normal distribution](#); however, it is more common to use the [inverse gamma distribution](#) formulation instead. This distribution is the [maximum entropy](#) distribution for a fixed first inverse [moment](#) ($E(1/X)$) and first logarithmic moment ($E(\ln(X))$).

Characterization

[edit]

The [probability density function](#) of the scaled inverse chi-squared distribution extends over the domain $x > 0$ and is

$$f(x; \nu, \sigma^2) = \frac{(\sigma^2 \nu / 2)^{\nu/2}}{\Gamma(\nu/2)} \frac{\exp\left[-\frac{\nu \sigma^2}{2x}\right]}{x^{1+\nu/2}}$$

where ν is the [degrees of freedom](#) parameter and σ^2 is the [scale parameter](#). The cumulative distribution function is

$$F(x; \nu, \sigma^2) = \Gamma\left(\frac{\nu}{2}, \frac{\sigma^2 \nu}{2x}\right) / \Gamma\left(\frac{\nu}{2}\right)$$

$$= Q\left(\frac{\nu}{2}, \frac{\sigma^2 \nu}{2x}\right)$$

where $\Gamma(a, x)$ is the [incomplete Gamma function](#), $\Gamma(x)$ is the [Gamma function](#) and $Q(a, x)$ is a [regularized Gamma function](#). The [characteristic function](#) is

$$\varphi(t; \nu, \sigma^2) = \frac{2}{\Gamma(\frac{\nu}{2})} \left(\frac{-i\sigma^2 \nu t}{2}\right)^{\frac{\nu}{4}} K_{\frac{\nu}{2}}(\sqrt{-2i\sigma^2 \nu t}),$$

where $K_{\frac{\nu}{2}}(z)$ is the modified [Bessel function](#) of the second kind.

Parameter estimation

[edit]

The [maximum likelihood](#) estimate of σ^2 is

$$\sigma^2 = n / \sum_{i=1}^N \frac{1}{x_i}.$$

The [maximum likelihood](#) estimate of $\frac{\nu}{2}$ can be found using [Newton's method](#) on:

$$\ln\left(\frac{\nu}{2}\right) + \psi\left(\frac{\nu}{2}\right) = \sum_{i=1}^N \ln(x_i) - n \ln(\sigma^2),$$

Scaled-inverse-chi-squared

Probability density function	None uploaded yet
Cumulative distribution function	None uploaded yet
Parameters	$\nu > 0$ $\sigma^2 > 0$
Support	$x \in (0, \infty)$
PDF	$\frac{(\sigma^2 \nu / 2)^{\nu/2}}{\Gamma(\nu/2)} \frac{\exp\left[-\frac{\nu \sigma^2}{2x}\right]}{x^{1+\nu/2}}$
CDF	$\Gamma\left(\frac{\nu}{2}, \frac{\sigma^2 \nu}{2x}\right) / \Gamma\left(\frac{\nu}{2}\right)$
Mean	$\frac{\nu \sigma^2}{\nu - 2}$ for $\nu > 2$
Mode	$\frac{\nu \sigma^2}{\nu + 2}$
Variance	$\frac{2\nu^2 \sigma^4}{(\nu - 2)^2(\nu - 4)}$ for $\nu > 4$
Skewness	$\frac{4}{\nu - 6} \sqrt{2(\nu - 4)}$ for $\nu > 6$
Ex. kurtosis	$\frac{12(5\nu - 22)}{(\nu - 6)(\nu - 8)}$ for $\nu > 8$
Entropy	$\frac{\nu}{2} + \ln\left(\frac{\sigma^2 \nu}{2} \Gamma\left(\frac{\nu}{2}\right)\right) - \left(1 + \frac{\nu}{2}\right) \psi\left(\frac{\nu}{2}\right)$
MGF	$\frac{2}{\Gamma(\frac{\nu}{2})} \left(\frac{-\sigma^2 \nu t}{2}\right)^{\frac{\nu}{4}} K_{\frac{\nu}{2}}(\sqrt{-2\sigma^2 \nu t})$
CF	$\frac{2}{\Gamma(\frac{\nu}{2})} \left(\frac{-i\sigma^2 \nu t}{2}\right)^{\frac{\nu}{4}} K_{\frac{\nu}{2}}(\sqrt{-2i\sigma^2 \nu t})$

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Probability density function [edit]

The inverse gamma distribution's [probability density function](#) is defined over the [support](#) $x > 0$

$$f(x; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} \exp\left(-\frac{\beta}{x}\right)$$

with [shape parameter](#) α and [scale parameter](#) β .^[1] Here $\Gamma(\cdot)$ denotes the [gamma function](#).

Unlike the [Gamma distribution](#), which contains a somewhat similar exponential term, β is a scale parameter as the distribution function satisfies:

$$f(x; \alpha, \beta) = \frac{f\left(\frac{x}{\beta}; \alpha, 1\right)}{\beta}$$

Cumulative distribution function [edit]

The [cumulative distribution function](#) is the [regularized gamma function](#)

$$F(x; \alpha, \beta) = \frac{\Gamma\left(\alpha, \frac{\beta}{x}\right)}{\Gamma(\alpha)} = Q\left(\alpha, \frac{\beta}{x}\right)$$

where the numerator is the upper [incomplete gamma function](#) and the denominator is the [gamma function](#). Many math packages allow you to compute Q, the regularized gamma function, directly.

Characteristic function [edit]

$K_\alpha(\cdot)$ in the expression of the [characteristic function](#) is the modified [Bessel function](#) of the 2nd kind.

Properties [edit]

For $\alpha > 0$ and $\beta > 0$,

$$\mathbb{E}[\ln(X)] = \ln(\beta) - \psi(\alpha)$$

	0	0.5	1	1.5	2	2.5	3
Parameters	$\alpha > 0$ shape (real) $\beta > 0$ scale (real)						
Support	$x \in (0, \infty)$						
PDF	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} \exp\left(-\frac{\beta}{x}\right)$						
CDF	$\frac{\Gamma(\alpha, \beta/x)}{\Gamma(\alpha)}$						
Mean	$\frac{\beta}{\alpha - 1}$ for $\alpha > 1$						
Mode	$\frac{\beta}{\alpha + 1}$						
Variance	$\frac{\beta^2}{(\alpha - 1)^2(\alpha - 2)}$ for $\alpha > 2$						
Skewness	$\frac{4\sqrt{\alpha - 2}}{\alpha - 3}$ for $\alpha > 3$						
Ex. kurtosis	$\frac{30\alpha - 66}{(\alpha - 3)(\alpha - 4)}$ for $\alpha > 4$						
Entropy	$\alpha + \ln(\beta\Gamma(\alpha)) - (1 + \alpha)\Psi(\alpha)$						
MGF	Does not exist.						
CF	$\frac{2(-i\beta t)^{\frac{\alpha}{2}}}{\Gamma(\alpha)} K_\alpha\left(\sqrt{-4i\beta t}\right)$						

Example: Consider again the measures of weighs (in g.) of the same object, performed with a precision balance: 2.64, 2.67, 2.65, 2.67, 2.65

We want to estimate the true weigh μ and the precision of the balance σ .

The inference for μ is the same using Bayesian or classical methods.

$\mu = 2.656 \pm 0.01664$ (check this!)



Of course, the interpretation of this 95% credibility (confidence) interval is different for Bayesian or classical inference.

To make inference for σ^2 , some information on the scaled inverse- χ^2 density will be needed. The most important is the following: **If X follows a $\text{inv-}\chi^2(\text{df}, k)$ then $\text{df} \cdot k / X$ follows a $\chi^2(\text{df})$.**

From here, we can see that the confidence intervals are “the same” using Bayesian or classical methods: $\sigma^2 = [0.00006, 0.00148]$ (check this!).



However, to find some point estimates we need to know the expectation as well.

Consequently, we obtain the following point estimates of the parameter σ^2 :

1- Expectation	0.0003591	$(n-1)s^2/(n-3)$
2- Mode	0.0000197	$(n-1)s^2/(n+1)$
3- Median	0.0002140*	
4- Classical (non-Bayesian) : $s^2 = 0.0001796$		

(*): $4*s^2/qchisq(0.5,4)$

Exercise: Make Bayesian inference for σ^2 using the measures of weighs.

- Assume that the true weigh ($\mu=2.6$) is known and we want to calibrate the balance, that is, we want to know σ^2 .
- Use the noninformative prior $\pi(\sigma^2) \sim 1/\sigma^2$
- Calculate point estimates and a 95% credibility interval.



Case study. There are many radiation accidents

- Radiotherapy accidents
- Manipulation of radioactive components in laboratories
- Industrial radiographies
- Disasters (Chernobyl (1986), Fukushima (2011))



From: Igor Kostin

There are many radiation accidents

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Accidente de Acerinox

El **Accidente de Acerinox** fue un incidente de [contaminación radiactiva](#) en [Cádiz](#) ([España](#)). En mayo de [1998](#), una fuente de [cesio-137](#) logró pasar a través del equipo de monitoreo de una planta de procesamiento de [chatarra](#) de [Acerinox](#) en [Los Barrios](#), [España](#). Cuando la chatarra fue derretida, el cesio-137 ardió y causó una nube radiactiva. Los detectores en la chimenea de la planta de Acerinox fallaron en detectarla, pero fue detectada en [Francia](#), [Italia](#), [Suiza](#), [Alemania](#) y [Austria](#). Los niveles de radiactividad que se midieron eran 1000 veces más altos que lo normal.

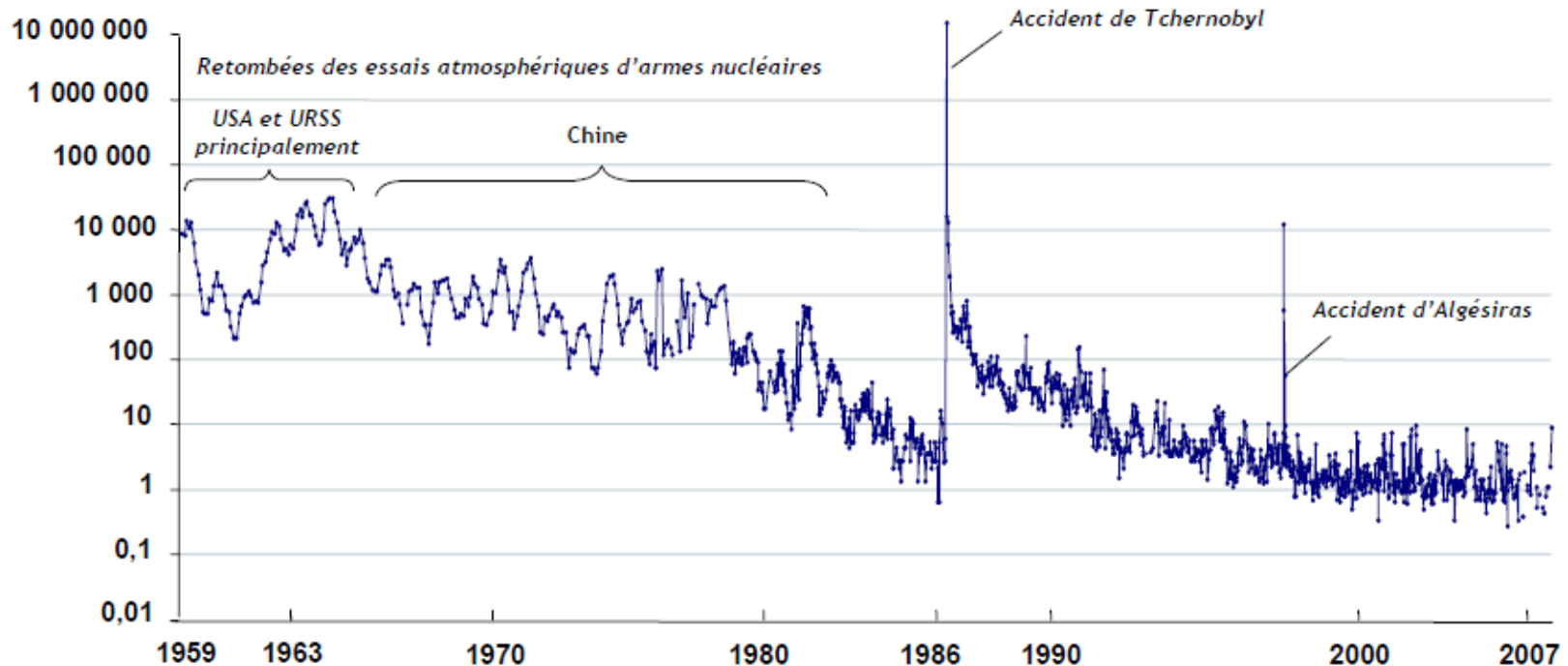
El accidente contaminó la planta de procesamiento de metal, más dos molinos de acero donde se enviaba basura para descontaminación. De acuerdo con laboratorios independientes,¹ las cenizas producidas por la fábrica de Acerinox tenían entre 640 y 1420 [becquerels](#) por gramo (la norma de [Euratom](#) es 10 Bq/g), suficientemente alto para ser considerado un peligro para el público.

Tras el accidente se hicieron revisiones médicas urgentes a todo el equipo que trabajaba en esa zona para comprobar si esos empleados estaban contaminados por el cesio.



Historique des mesures de césium 137 dans l'air en France depuis le début des années 60

Activité volumique du ^{137}Cs dans les aérosols $\mu\text{Bq}/\text{m}^3$ d'air



Station OPERA de la Seyne-sur-Mer

Natural radiation also exists

Banana equivalent dose

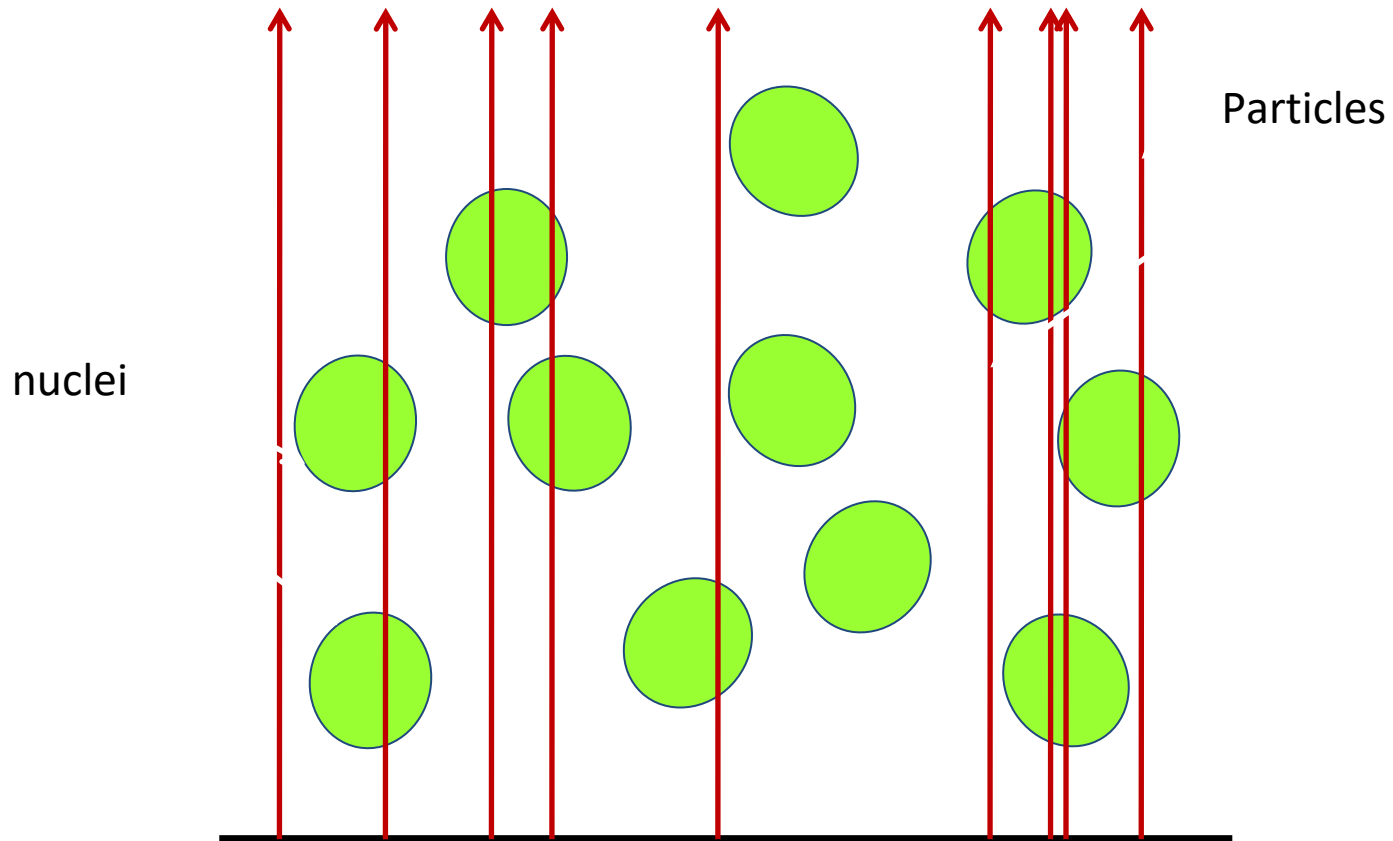
From Wikipedia, the free encyclopedia

A **banana equivalent dose (BED)** is an informal expression of [ionizing radiation](#) exposure, intended as a general educational example to indicate the potential dose due to naturally occurring [radioactive isotopes](#) by eating one average-sized [banana](#). One BED is often taken as 0.1 μSv , however, in practice this dose is not [cumulative](#), as the principal radioactive component is excreted to maintain [metabolic equilibrium](#). The BED is only an indicative concept meant to show the existence of very low levels of natural radioactivity within a natural food and is not a formally adopted dose quantity.

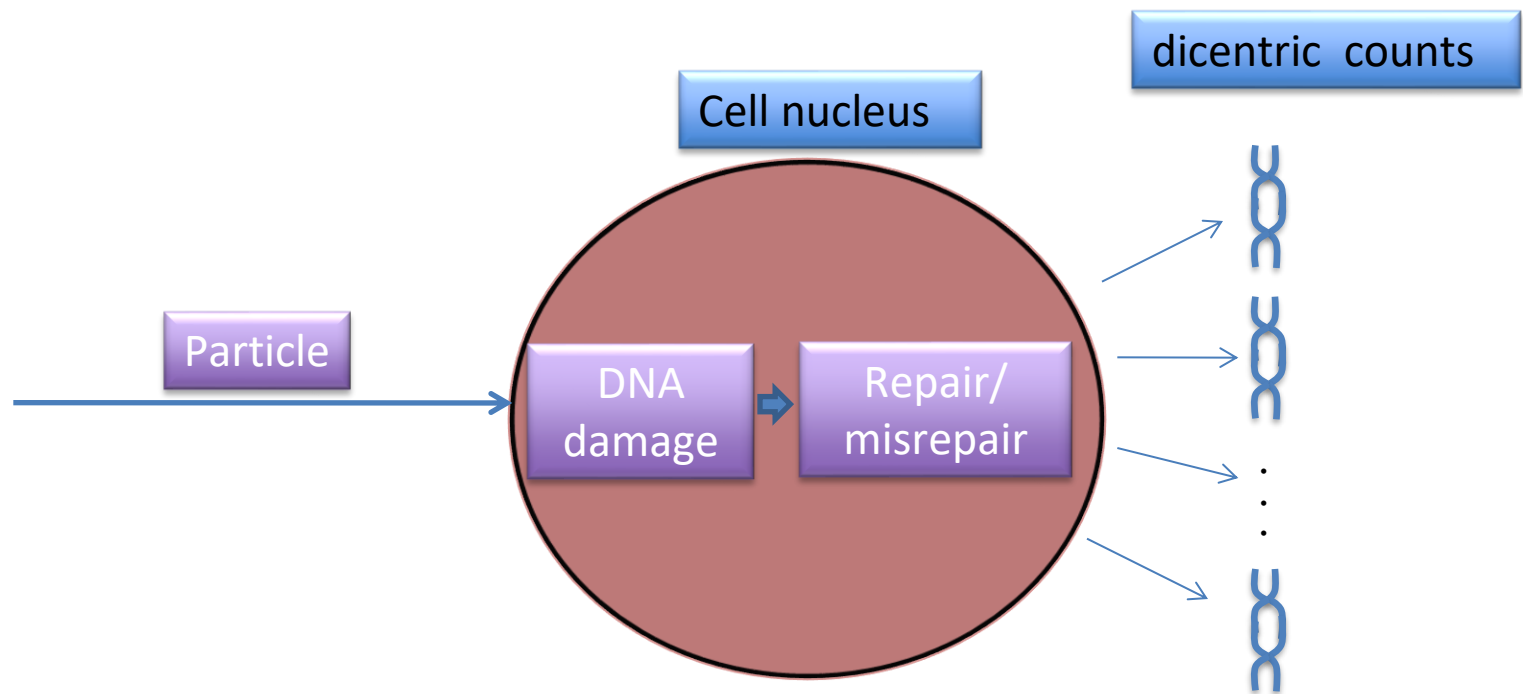
For example, the radiation exposure from consuming a banana is approximately 1% of the average daily exposure to radiation, which is 100 banana equivalent doses (BED). The maximum permitted radiation leakage for a nuclear power plant is equivalent to 2,500 BED (250 μSv) per year, while a chest [CT scan](#) delivers 70,000 BED (7 mSv). A [lethal dose](#) of radiation is approximately 35,000,000 BED (3500 mSv). A person living 10 miles from the [Three Mile Island](#) nuclear reactor received an average of 800 BED of exposure to radiation during the 1979 accident.^[1]



A banana contains [naturally occurring radioactive material](#) in the form of [Potassium-40](#). 



Poisson-distributed events. The figure shows events, one event is a track (in red) intersecting a cell nucleus (green)



Dicentric formation as a Compound Poisson process

The main goal: what is the dose received ?



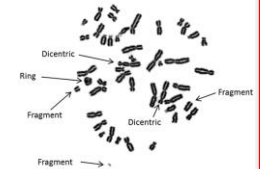
Irradiated patient



Blood sample



Cytogenetic methods



Frequencies of chromosome aberrations in peripheral lymphocytes

OBSERVED	cells with X dicentrics									CELLS	DIC	Freq
	0	1	2	3	4	5	6	7	8			
224	180	72	19	4	1	0	0	0	0	500	402	0,80

Quantification of the radiation dose received

This is essential for predicting the derived health consequences, for both early effects (vomiting, skin injuries, etc.) and late effects like the development of cancer.

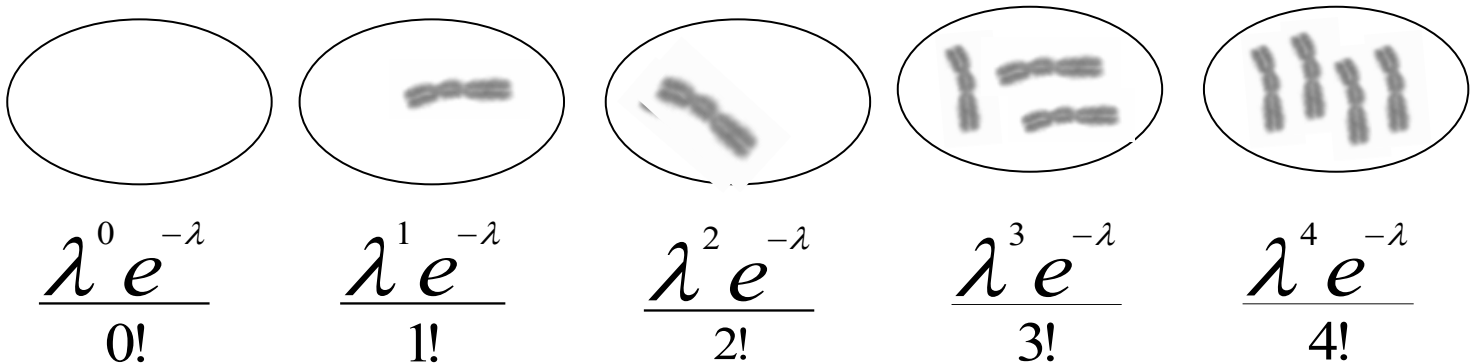
The most widely used method is the analysis of the frequency of chromosome aberrations observed in metaphases from a sample of peripheral blood lymphocytes.

How the aberrations are distributed among cells

The Poisson distribution is a **discrete probability distribution** that expresses the probability of a number of events occurring in a fixed period of time if these events occur with a known average rate and independently of the time since the last event.

If the expected number of occurrences in this interval is λ , then the probability that there are exactly n occurrences ($n = 0, 1, 2, \dots$) is equal to:

$$P(X = n) = \frac{\lambda^n e^{-\lambda}}{n!}$$

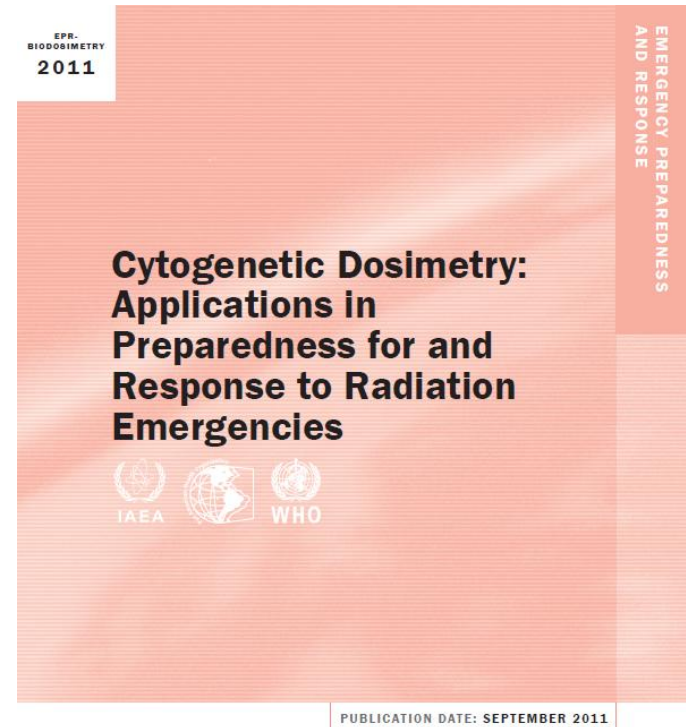


λ is related to the dose received: it increases when the doses increases

According to the IAEA manual:

The Poisson distribution is by far the most widely recognized and commonly used type of distribution for cytogenetic data analysis. (p. 197)

For dicentrics, irradiation with X or gamma rays (low LET radiation) produces a distribution of damage which is very well represented by the Poisson distribution.



Bayesian dose estimation

Suppose that we have a potentially irradiated patient and we have observed 64 dicentrics in 104 cells. What has been the received dose?

The first step is to do a calibration. Calibration can be experimentally difficult according to the kind of radiation.

doses (Gy) =	0.5	0.75	1.0	1.5	2.0	2.5	3.0
Number of cells=	269	78	115	90	84	59	37
Number of dicentrics=	109	47	94	114	138	125	97

(Groer and Pereira, 1987)

A Poisson model is usually assumed: $z_i \sim \text{Poisson}(n_i \alpha d_i)$ (lineal model)

where z_i indicates the number of dicentrics, n_i the number of cells, d_i the doses and α is a unknown parameter.

Now the likelihood is given by, $L(Z|\alpha) \propto e^{-\alpha \sum_{i=1}^n n_i d_i} \alpha^{\sum_{i=1}^n z_i}$

The prior $\pi(\alpha) \approx \text{Gamma}(a,b)$ is reasonable because this is a conjugate prior for the Poisson distribution.

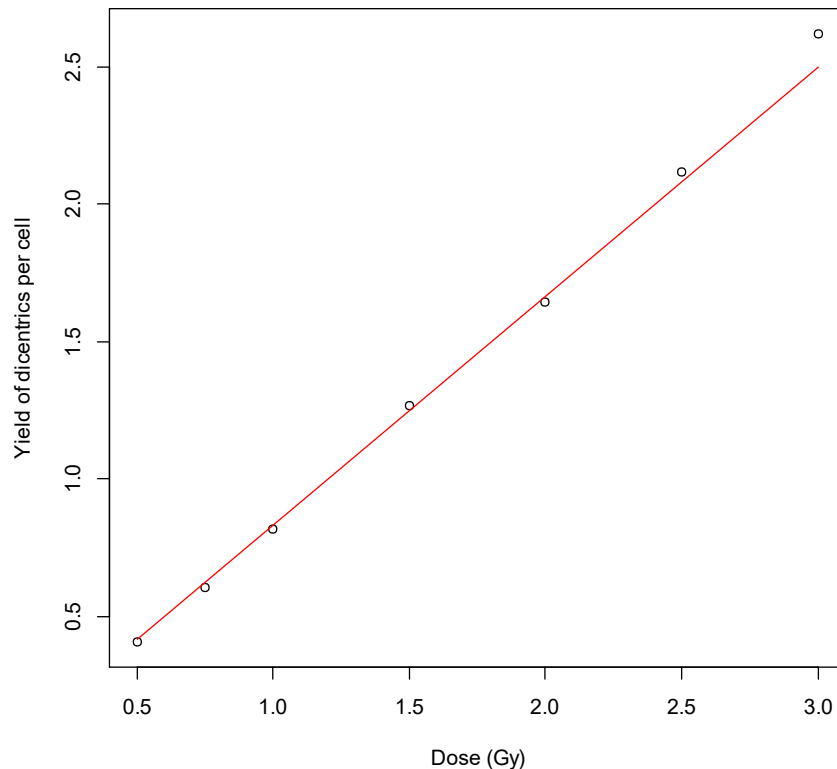
Then, the posterior distribution is, $f(\alpha|Z) \approx \Gamma(a + \sum_{i=1}^n z_i, b + \sum_{i=1}^n n_i d_i)$

How to choose the hyperparameters a and b?

- Taking into account the information from other laboratories.
- Published papers.
- Perhaps being redundant (using the own data).

The classical (non Bayesian) procedure

Calibration data and fitted line



The MLE estimator can be easily calculated:

$$\hat{\alpha} = \frac{\sum_{i=1}^k z_i}{\sum_{i=1}^k n_i d_i} \quad \hat{\sigma}(\hat{\alpha}) = \frac{\sqrt{\sum_{i=1}^k z_i}}{\sum_{i=1}^k n_i d_i}$$

With an observed value of $y=64/104=0.61$ an inverse regression analysis could be performed.

In this case,

$$\hat{\alpha} = 0.833, \quad \hat{\sigma}(\hat{\alpha}) = 0.031$$

Therefore, the use of a gamma distribution as a prior for α , with $a=722$ and $b=867$, would agree with the MLE of α and its standard deviation estimate.

For this data the posterior distribution would be, $f(\alpha|Z) \approx \Gamma(1446, 1736.5)$

Now we have newly collected data in the form of y total aberrations in n_y cells, for dose d , represented by a Poisson distribution:

$$p(y|\alpha, d) = \frac{e^{-\alpha n_y d} (\alpha n_y d)^y}{y!}$$

In this case, $y=64$ and $n_y=104$.

We assume now a prior distribution of the form, $\pi(\alpha, d) = \pi(\alpha) \pi(d)$, that is, α and d are independent. Accordingly with the nature of the accident, Groer and Pereira (1987) use $\pi(d) \approx \text{Gamma}(10, 10)$. It is reasonable to take as a prior for α the posterior obtained from the calibration experiment, that is, $\pi(\alpha) \approx \text{Gamma}(1446, 1736.5)$. Consequently, the new posterior is,

$$f(\alpha, d|y) \propto e^{-104\alpha d} (\alpha d)^{64} \alpha^{1446-1} e^{-1736.5\alpha} d^{10-1} e^{-10d}$$

But we are only interested in the marginal posterior distribution of d , that is,

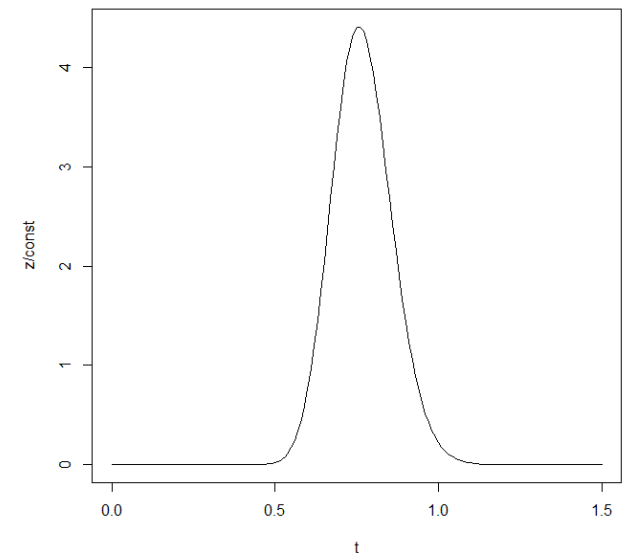
$$f(d|y) = \int_0^\infty f(\alpha, d|y) d\alpha \propto d^{74-1} e^{-10d} \int_0^\infty \alpha^{1510-1} e^{-(1736.5+104d)\alpha} d\alpha$$

$$\propto \frac{d^{73} e^{-10d}}{(1736.5+104d)^{1510}}$$

This is not a Gamma distribution, it is a different density.

Plotting the density:

```
t=seq(0,1.5,.01)
postapprox=function(d){exp(
-10*d+73*log(d)-1510*log(1736.5+104*d)+11380)}
z=postapprox(t)
const=integrate(postapprox,0, Inf, rel.tol=1.0)$value
plot(t,z/const,type="l")
```



This is the mean:

```
me=function(d){exp(
-10*d+74*log(d)-1510*log(1736.5+104*d)+11380)/const}
integrate(me,0, Inf, rel.tol=1.0)$value
```

0.768 Gy

A 95% credibility interval:

```
dens=function(d){exp(
-10*d+73*log(d)-1510*log(1736.5+104*d)+11380)/const}
dist=function(u){integrate(dens,0, u)$value}
uniroot(function(u){dist(u)-0.025},c(0,.7))
uniroot(function(u){dist(u)-0.975},c(.7,1.5))
```

The interval is (0.600, 0.956) Gy

EFECTOS DETERMINISTAS: NIVEL TISULAR

Tejido	Efecto	Periodo de latencia aproximado	Umbral aproximado (Gy)	Dosis efectos severos	Causa
Sistema hematopoyético	Infecciones Hemorragias	2 semanas	0,5	2,0	Leucopenia Plaquetopenia
Sistema Inmune	Inmunosupresión Infección sistémica	Algunas horas	0,1	1,0	Linfopenia
Sistema gastrointestinal	Deshidratación Desnutrición	1 semana	2,0	5,0	Lesión del epitelio intestinal
Piel	Escamación	3 semanas	3,0	10,0	Daño en la capa basal
Testículo	Esterilidad	2 meses	0,2	3,0	Aspermia celular
Ovario	Esterilidad	< 1 mes	0,5	3,0	Muerte interfásica del oocito
Pulmón	Neumonía	3 meses	8,0	10,0	Fallos en la barrera alveolar
Cristalino	Cataratas	> 1 año	0,2	5,0	Fallos en la maduración
Tiroides	Deficiencias metabólicas	< 1 año	5,0	10,0	Hipotiroidismo
Sistema nervioso central	Encefalopatías y mielopatías	Muy variable según dosis	15,0	30,0	Demielinización y daño vascular